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4 / 4 submitted

Level 2 — 5 marks/question

L2-Q1. (M) Frequency

p

CODE

```
a = eval(input())

freq = {}

for x in a:
    if x in freq:
        freq[x] += 1
    else:
        freq[x] = 1
        print("Element | Frequency")
        print("-----")

    for x in freq:
        print("x,      |",freq[x])

    m = max(freq.values())
    for x in freq:
        if freq in x == [m]:
            print("output", x, "&", m)
```

OUTPUT

Security Error: Disallowed code pattern detected.

L2-Q2. (M) 2D Multiplication

p

CODE

```
a = [[1, 2], [5,3]]
b = [[2,3], [4,1]]
c = [[sum(a[i][k] * b[k][j] for k in range (2)) for j in range (2)]for i in range(2)]

for row in c:
    print(*row)
```

INPUT

```
1 2
5 3
2 3
4 1
```

OUTPUT

```
10 5
22 18
```

L2-Q3. (M) Words-Lines

P

CODE

```
import sys
lines = sys.stdin.read().splitlines()
print("No. of words in lines =",len(lines[0].split()))
print("No. of lines in files =",len(lines))
```

INPUT

"Machine learning (ML) is a kind of AI (artificial intelligence) that lets software applications become more precise at anticipating results without being specifically programmed to do so. Machine learning algorithms utilize historical data to anticipate/predict the latest output value s."

OUTPUT

No. of words in lines = 13
No. of lines in files = 4

L2-Q4. (M) Lucky

P

CODE

```
def rsum(n):
    if n == 0:
        return 0
    return n % 10 + rsum(n // 10)

def rprod(n):
    if n < 10:
        return n
    return n % 10 * rprod(n // 10)

n = (int(input()))

if rsum(n) == rprod(n):
    print("YES")
else:
    print("NO")
```

8 / 60

INPUT

1412

OUTPUT

YES